Title:

Leaving the most vulnerable behind? School closures during the COVID-19 pandemic and pre-existing life struggles in Bolivia

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Abstract

School closures were widely applied to curtail the spread of the coronavirus. By mid 2020, 1.3 billion learners were out of school, and many governments implemented virtual learning modalities. Yet a large number of students were ill-prepared to effectively adopt it. In Bolivia, many learners lacked internet access and technological devices, while schools remained closed for over 41 weeks since March 2020. These deprivations regularly overlap with limited income, poor health, and deficient dwelling infrastructure, among others. This situation implies increased risks of school dropout, particularly for children living in vulnerable households. We combine microsimulation and projection methods to assess the short- and mid-term effects of school closures on school unattendance in Bolivia. We show that the proportion of people living in households affected by school unattendance may increase from 2.1% to 8.5% if the 10% most vulnerable children drop out of school. We also analyze the joint distribution of school unattendance and additional livelihood deprivations. The newly acquired school attendance deprivations are combined with pre-existing life struggles pushing people into multidimensional poverty, understood as the experience of simultaneous hardships. We thus gauge how school closures may have hindered progress made towards ending poverty in all its forms and dimensions.

Keywords: School Closures; COVID-19; School Unattendance; Microsimulations; Projections; Overlapping Deprivations

JEL Code: I25; I32

1. Introduction

Before the eruption of the COVID-19 global pandemic in early 2020, steady and significant improvements in many development indicators had been taking place. Many countries in the developing world were moving towards universal enrollment in primary education (Friedman et al., 2020); there was marked progress in reducing monetary poverty over the past decades (World

Bank, 2018), and important reductions of overlapping livelihood deprivations in the domains of health, education, and living standards were anticipated across some of the poorest countries in the world (Alkire et al., 2020). However, the pandemic has put this progress at risk. In order to reduce the spread of the coronavirus, drastic policy responses in 2020 and 2021 were applied globally; they consisted of very restrictive measures, including full school closures as part of country-wide lockdowns (UNESCO, 2020a). As a result, more than 1.2 billion students at all levels of education worldwide had stopped having face-to-face classes by mid-2020 (ECLAC-UNESCO, 2020). Most countries turned to virtual learning in an attempt to offset the potential negative consequences of such an unprecedented measure on school attendance and overall stability of learning processes. However, many learners in developing countries were ill-prepared to swiftly adapt to digital learning formats due to the several simultaneous disadvantages that they regularly face, going from lack of connectivity to severe economic handicaps (ECLAC-UNESCO, 2020; Gomez and Uzin, 2022).

Crucially, school closures give rise to serious risks of school dropouts. Evidence shows that disruptions to the schooling process may increase the opportunity cost of staying in school, especially for households facing financial stress. In the 1916 polio epidemic, researchers hypothesize that children of legal working age (13 in most U.S. states at that time) were more likely to leave school permanently following epidemic-related shutdowns (Azevedo et al., 2021). Likewise, evidence from school closures caused by the Ebola pandemic in Guinea and Sierra Leone shows that youth in the poorest households saw the largest increase in school dropout, exceeding expected pre-Ebola dropout rates[1] (Smith, 2021). There is also evidence from Chile showing that, between 2000 and 2012, school closure increased the probability of high-school dropout between 46 and 62 percent (1.7 and 2.3 percentage points, see Grau et al., 2018). Therefore, increased risks of school absenteeism and subsequent dropout can be anticipated as a consequence of the 2020 pandemic. Moreover, these risks may be higher in countries where the length of school closures was greater. Evidence from the Ebola pandemic reveals that the pandemic did more harm to the school dropout rate in Sierra Leone, where there was a longer school closure (9 months), than in Guinea (7 months) (Smith, 2021). Indeed, Azevedo et al. (2021) compellingly show that longer school closures during the 2020 pandemic may cause greater learning loses, as well as prospects of higher school dropout rates.

Bolivia, the poorest country in South America, stands out as a particularly relevant case study in this context because, despite considerable efforts made by the Ministry of Education to widely promote virtual learning since March 2020, it was impossible to include the entire student population in this process due to lack of universal internet connectivity, limited access to digital devices, and lack of digital skills among learners and teachers – particularly in the rural areas (IDB, 2020). According to UNICEF (2020), four out of ten learners in Bolivia were not able to attend any class in 2020. They also report that 93.2 percent of students who were attending virtual classes were learning "nothing", "almost nothing" or "more or less", regardless of the

[1] An additional 17,400 of the poorest secondary-age youth were out of school.

type of school they were enrolled in (private or public). While online education was possible for some students, it is clear that the overall quality of distance learning was suboptimal. This situation poses great threats to sustainable development in the country on its own, but importantly, the pre-pandemic socioeconomic context in Bolivia exhibits several structural deprivations and inequalities that aggravate the situation for some students (World Bank, 2020a; OPHI, 2021). Moreover, Bolivia is currently among the countries that have decided to keep schools fully or partially closed the longest (92 weeks), increasing the risk of leaving the most vulnerable behind (UNESCO, 2020a; Gómez and Uzín, 2022).

In this paper we stress that in order to adequately gauge the effects of school closures, it is important to understand that they often imply an *additional* livelihood deprivation. It is well documented in the development literature that households lacking the means to keep children in school also suffer from several other life struggles simultaneously, such as deprivations in health (Grossman, 2015; Bratti and Mendola, 2014), in parental education (Klein et al., 2020), and housing infrastructure (Momo et al., 2021; Pillay, 2017). We argue that it is very likely that students struggling to adapt themselves to virtual learning face similar overlapping deprivations. Thus, a framework that assesses disruptions in school attendance while also accounting for other living conditions is required to truly grasp the matter at hand. One such framework is the dual cutoff counting approach to measure overlapping deprivations developed by Alkire and Foster (2011). This framework underpins several analyses of simultaneous life struggles generally – such as UNDP's global Multidimensional Poverty Index (Alkire and Santos, 2014) – and it has also been applied to assess the *aggregate* effects of school closures during the pandemic in a set 75 developing countries (Alkire et al., 2021a). We thus use the dual cutoff framework to contribute to the continuing public debate on policy responses to COVID-19 by offering a household-level analysis of the effects of school closures in Bolivia through the lens of a multidimensional approach to assess risks of school unattendance alongside a set of additional pre-existing deprivation indicators.

We argue that in spite of clear limitations, including data scarcity and the ensuing impossibility of proper impact evaluations, policy action needs to take place, but it needs to build on the most objective information available. Hence, this paper overcomes this particular data limitation by applying a Monte Carlo microsimulation approach. Household-specific school unattendance risks are simulated using economic and digital vulnerability measures. We simulate the potential impact of the pandemic identifying students that are likely to struggle adapting to virtual learning due to lack of appropriate technological equipment and internet connectivity and drawing inspiration from Brown et al., (2020) and Azevedo et al., (2021), we build a dropout vulnerability score drawing that includes access to internet, ownership of computer, mobile phones, radio, and television. Additionally, a heterogeneity analysis is performed to uncover subnational specificities.

Importantly, in the case of developing countries, such as Bolivia, there are high risks of disrupting sustainable development processes for children today and in the future if policy action

is based on thin evidence. We thus aim at investigating not only the immediate effects of school closures, but also the extent to which they may hinder processes of sustainable development by increasing risks of missing development targets that were otherwise achievable. To do this, we identify the dynamic trajectories of i) school attendance rates in Bolivia accounting for risks of school dropout due to unsuccessful adaptation to virtual learning, and ii) aggregate measures of the joint distribution of these school unattendance risks and additional pre-existing livelihood deprivations.

Among our most salient results, we find that school closures may have easily increased the observed incidence rates of deprivations in school attendance by four times. For instance, if 10% of households that are most likely to become deprived in school attendance become effectively deprived, the proportion of people living in households where there is at least one school-aged child out of school would have gone up from 2.1% to 8.5%. The simulated additional deprivations in school attendance combine themselves with pre-existing livelihood deprivations in other indicators included in the global MPI in such a way that they can push a considerable number of people into multidimensional poverty. Under the same scenario, the proportion of people in multidimensional poverty would rise from 11.5% pre-pandemic to more than 14%. This implies that the proportion of people that are multidimensionally poor and deprived in school attendance would then go from 1.3% to 7%.

The paper is structured as follows. Section 2 presents a brief literature overview, with emphasis on evidence about the overlap between school unattendance and additional livelihood deprivations. Section 3 presents our methods in detail. First, we explain our microsimulation approach to measure the effects of school closures on school unattendance. We also explain how we measure the joint distribution between school unattendance and pre-existing deprivations in the domains of health, education, and living standards – this framework follows the UNDP-OPHI global Multidimensional Poverty Index (MPI) structure and methods. Finally, we explain how we project school unattendance and overlapping deprivation measures into 2030 to assess the effect of school closures on a 2030 Agenda-inspired discussion. Section 4 presents our main results, and Section 5 presents some final remarks.

2. Literature overview

2.1. School closures due to the pandemic and policy responses

The COVID-19 pandemic has posed unprecedented threats to education in more than one way. On the one hand, it directly affects learning continuity through the immediate impact of school closures around the world. In 2020, the pandemic has led to massive closures of face-to-face activities of educational institutions in more than 190 countries. By early 2022, partial school closures persisted in many countries, particularly in the developing world where schools remained closed the longest (UNESCO, 2020a). The main consequences of this effect include learning losses, increased school absenteeism, more dropouts, and higher inequality (World Bank, 2020b). For instance, using data for 157 countries, Azevedo et al. (2021) simulated the

effect of the pandemic on schooling and found that COVID-19 could result in a loss of between 0.3 and 0.9 years of schooling adjusted for quality. Similarly, according to the UNESCO Responses to Educational Disruption Survey (REDS) report covering 11 countries, many teachers who were engaged in virtual classes during 2020 reported a decrease in student attendance (ranging from 25% in India to 85% in Uruguay), and more than 50 percent of teachers reported decreases in student learning outcomes. Many school principals stated that learning outcomes of disadvantaged students – by past academic records – decreased more than those of their better-off peers during the pandemic (ranging from 23% in Uzbekistan to 79% in Uruguay) (UNESCO-IEA, 2022).

On the other hand, the pandemic also affected education indirectly through the related economic crisis, which jeopardized key drivers of learning. For one, education supply can still be expected to suffer a contractive shock due to potential cuts in education public investments (World Bank, 2020b). Moreover, the demand for education is also likely to decrease due to more severe household budget constraints to keep their children out of school. For instance, Azevedo et al. (2021) find that close to 7 million students from primary and secondary education could drop out due to the pandemic-related income shock. This is particularly true for the most vulnerable households. In fact, according to the World Bank (2020b), the most educated and wealthiest families are expected to have sufficient equipment, connectivity, and knowledge to cope with the challenges posed by the pandemic and sustain their children's learning at home. Meanwhile, students facing several additional deprivations could be especially vulnerable to school absenteeism and dropout.

As the pandemic continues, full reopening of schools is unlikely to happen in the short term in many developing countries. If disruptions to the learning process persist, we are increasingly likely to see permanent negative effects on human capital accumulation and development. Initial policy responses in most countries around the developing world consisted of adopting remote education formats, including the set-up of multimodal TV and radio channels, and promoting digital competencies among teachers to support remote learning. Monitoring processes to evaluate the effectiveness of remote education programmes were implemented in Brazil, Kenya, Nigeria, Peru, and Sierra Leone (Barron-Rodriguez et al., 2021). According to an evaluation conducted by Fundaçao Lemann (2020), remote learning programs in most Brazilian states only reached 74 percent of the student population, and access stability was highly unequal – while 94 percent of students in the southern region enjoyed stable access to remote learning resources, that was the case of only 52 percent of students in the northwest region. Similarly, according to Uwezo (2020), only 22 percent of school-going children in Kenya had access to digital resources for remote learning, and students in private schools were twice as likely to access remote learning than students in public schools. Additionally, in the Nigerian state of Edo, only 29 percent of children in primary school accessed the government's interactive mobile-based platform. In Peru over 85 percent of students accessed the remote learning program *Aprendo en Casa*, and in Sierra Leone, almost 80 percent of the population reported that children in their households were

spending less time on education activities compared to the pre-pandemic situation (Munoz-Najar and Oviawe, 2022; Munoz-Najar, A., 2022; IPA, 2020).

Chiefly focused on developed countries, a growing strand of literature is documenting the negative effects of school closures on learning outcomes. For instance, it has been found that learning losses among Dutch students after a relatively short period of school closure (8 weeks) are equivalent to one-fifth of the school year (3 percentage points). Moreover, they are 60 percent larger among students with less-educated parents compared to more advantaged children in this respect (Engzell et al. 2020). In the Flemish region of Belgium, it was found that mathematics and language scores decreased by 0.19 and 0.29 standard deviations during the pandemic respectively for students in grade 6 compared to previous cohorts (Maldonado and De Witte 2020). Similarly, findings in that study also suggest that schools with a higher proportion of students coming from households with more privileged socioeconomic status tend to suffer lower learning losses from school closures. Shult et al. (2020) found that in Germany, children in primary education suffered from academic score losses of around 0.07 standard deviations for reading and 0.09 standard deviations for math in 2020 compared to previous years, and these reductions were more important for lower-achieving students.

This evidence suggests that even children in developed countries, where there is likely to be a technologically-advanced remote learning setup, made little or no progress on learning outcomes during the pandemic. Governments of some developing countries (such as the Caribbean and Latin American states, see Parker and Alfaro, 2022; and UNESCO, 2022b) provided technological devices as one of the measures adopted to implement distance learning activities. However, dealing with the potential schooling setback due to the pandemic is not just an issue of differences in access to equipment, or technological skills. Unequal access to remote learning opportunities widens pre-existing gaps that must be viewed from a multidimensional perspective. The World Bank (2020b), stressed the importance of considering different deprivation dimensions, including parental education to help their children sustain their engagement with (virtual) education, the disadvantages of not having a proper household infrastructure for remote learning, and ways to provide access to learning to students with disabilities, and other health issues. Precisely, our motivation in this paper is to highlight the importance of adopting a multidimensional perspective for education policy design and implementation in order to draw attention towards the most vulnerable population as this pandemic continues to unfold.

2.2. Overlapping deprivation assessment

Understanding the problem of school closures during the pandemic from a multidimensional perspective is key to developing an effective policy that guarantees the prioritization of vulnerable households. This is clearly aligned with the 2030 Agenda for Sustainable Development. Specifically, SDG 4 calls to "ensure inclusive and equitable quality education and to promote lifelong learning opportunities for all". Importantly, as vulnerability to disruption of the learning process in our context may be perceived as a monetary issue, income or expenditure

measures would usually be given high analytical prominence. However, access to income itself at the household level may not directly improve students' wellbeing because their needs may not necessarily be fulfilled by higher household incomes (De Neubourg et al., 2013). Thus, vulnerable groups in our context need to be identified considering multiple livelihood deprivations. Besides income or access to technology, children's ability to be (and remain) in school, even pre-pandemic, is far from being randomly distributed across the population. For instance, there is vast evidence on the negative effects of health deprivations within the household on the schooling of children. For instance, Alderman et al. (2001) found substantial effects of preschool nutrition on the likelihood of school enrollment in rural Pakistan. Jackson et al. (2011) found that children with poor oral health status in North Carolina were nearly 3 times more likely to skip school. Another important factor to consider is parental education, which may play an even more important role in virtual learning than in the pre-pandemic situation. Klein et al. (2020) found that social housing and parental education in Scotland had the most pervasive associations with school absences across all forms of absenteeism. Similarly, Chevalier et al. (2013) found greater effects of parental schooling on school dropout in the UK than those generated by parental income. In addition, households that are not capable to keep their children in school are likely to suffer from deprivations in dwelling infrastructure. Momo et al. (2021) find that a low household socioeconomic status[2] critically hinders the likelihood of upward intergenerational educational mobility. Finally, Pillay (2017) found that most learners who live in low-cost houses in South Africa (such as shacks) perform poorly in literacy tests compared to those learners who live in more conventional (brick) houses.

Clearly, a framework that considers additional living conditions is necessary to identify the most vulnerable groups to the negative effects of the pandemic on schooling. One such framework has been developed around the notion of 'multidimensional poverty', which considers various simultaneous deprivations that people experience including poor health, lack of education, and inadequate living standards (OPHI, 2019). Using microdata, the most frequently used methodology for measuring multidimensional poverty is the AF method (Alkire and Foster, 2011), which allows for identifying people experiencing a critical number of weighted deprivations and estimating aggregate measures of this condition. In 2010, based on the AF method, the global Multidimensional Poverty Index (gMPI) was first published by UNDP in the Human Development Report (United Nations Development Programme, 2010; Alkire and Santos, 2010, 2014), as an internationally comparable overlapping deprivation index based on the joint distribution of 10 indicators grouped under the dimensions of Health, Education, and Living Standards. Since its inception and structural update in 2018 (see Alkire et al., 2022), the gMPI has become a benchmark measure to gauge overlapping deprivations in academic and policy-making spheres. Several countries have developed national adaptations based on the global MPI (OPHI and Human Development Initiative, 2019), but studies focused on countries that do not have a national multidimensional poverty measure, which is the case of Bolivia, have

[2] Including cooking fuel of dung, wood or charcoal; unimproved or shared sanitation facilities; unsafe or shared drinking water; no electricity; unimproved housing; and absence of assets

commonly adopted the global MPI as an empirical framework to assess overlapping deprivations (see e.g. Dawas, 2018 for Jordan). This is precisely what we do in this paper.

3. Data and methods

3.1. Data

The data used in this study come from the 2013—2019 household surveys conducted annually by the Bolivia National Institute of Statistics (INE). These surveys are the only nationally representative, official source of information about sociodemographic, educational, housing and health characteristics[3] of Bolivian households. The timespan of our analysis is entirely defined by strict data and sampling comparability criteria across yearly surveys[4]. Combining all datasets, our study uses information coming from 64,884 household-level observations.

According to UNESCO (2020a), nearly three million learners in Bolivia were negatively affected by the school closures caused by the pandemic in 2020. They also report a total of 82-week duration of school closures due to the pandemic, which makes Bolivia one of the countries that have decided to keep schools fully or partially closed the longest (UNESCO, 2020a).

In this context, we are particularly interested in variables related to access to technological equipment, information technologies and internet services, which were necessary tools to carry out the distance learning modalities that were adopted in the year of the pandemic. We take this information from the 2019 Household Survey, which is the largest and best source of information at the pre-pandemic household level. Table 1 presents information regarding remote learning variables and presents the frequency and percentage of eligible households[5] reporting lack of access to different services in Bolivia in 2019. Note that lack of internet connectivity from cable at home was extremely high in 2019, as 78.3% of eligible households (around 2.6 million households) reported having no access to the internet. This problem is higher in the subnational regions of Potosi and Chuquisaca. Similarly, a high proportion of households did not have access to a computer (74.3%). Even if relatively few households had access to a radio (38.1%), a large number of people had access to a mobile telephone and TV (95.7% and 89.3%, respectively).

[3] These surveys are not specifically designed to collect data on health characteristics like the Demographic and Health Surveys (DHS). However, they do contain some health information.

[4] The complete description of the specifics can be found in http://anda.ine.gob.bo/index.php/catalog/ENCUESTAS

[5] Households that have at least one child attending school.

Table 1: Description of remote learning variables in Bolivia, 2019

Lack of access to…	Internet		Computer		Mobile telephone		Radio		TV	
	N	%	N	%	N	%	N	%	N	%
National	2,659,469	78.3	2,510,532	74.3	304,432	4.3	2,081,729	61.9	556,901	10.7
Chuquisaca	159,203	91.5	133,244	77.7	15,686	3.8	88,503	46.5	47,378	22.2
La Paz	723,308	82.9	626,061	73.5	59,576	0.7	391,423	45.9	104,377	6.8
Cochabamba	449,681	79.0	402,362	71.2	37,945	1.5	339,524	54.9	80,203	10.3
Oruro	144,974	80.9	124,139	66.4	9,694	0.8	90,397	56.7	40,926	14.0
Potosi	268,062	94.9	220,510	71.7	50,797	5.3	95,033	37.4	89,522	17.8
Tarija	133,822	76.3	133,385	76.4	14,466	4.6	110,402	64.7	24,040	9.2
Santa Cruz	621,010	65.1	714,327	74.5	85,090	6.3	804,843	82.6	128,431	9.0
Beni	124,447	89.4	123,530	89.9	26,125	18.7	121,414	85.8	32,593	19.7
Pando	34,963	82.4	32,974	78.9	5,053	11.4	40,190	97.3	9,431	17.6

Notes. N is obtained by weighting the number of observations in the survey by its expansion factor.

3.2. Methods

We aim at contributing to the continuing public debate on policy responses to COVID-19 by offering an analysis based on a multidimensional approach to assess threats to school attendance alongside an additional set of relevant livelihood indicators. In this section we describe the different methods to do so. First, in order to account for other living conditions, we specify the framework adopted, which is inspired by the global Multidimensional Poverty Index (MPI) (Alkire and Santos, 2014). Then, we describe the method used to identify the dynamic trajectories of the global MPI-related indicators to nowcast and project both school attendance ratios and other relevant socioeconomic characteristics. Finally, we explain our strategy to simulate school absence shocks, based on estimated dropout vulnerability scores for each household.

3.2.1. Multidimensional poverty measurement

As stated previously, the joint distribution of household deprivations is measured following the AF method (Alkire and Foster, 2011). Our multidimensional deprivation assessment builds from a matrix of achievements $X = [x_{ij}]$, where entry x_{ij} is the achievement of the household $i = 1, 2, \dots, n$ in indicator $j = 1, 2, \dots, d$. Indicator-specific cutoffs, z_j, denote the minimum accepted level of achievements for a household to be considered non-deprived. These cutoffs are presented in Table 2, and they allow us to transform the achievement matrix X into a deprivation matrix denoted as g^0 whose typical entry g_{ij}^0 is a binary indicator: $g_{ij}^0 = 1$ when $x_{ij} < z_j$ – deprived household, and $g_{ij}^0 = 0$ otherwise – non-deprived household.

In our setting, which mimics that of the UNDP-OPHI global multidimensional poverty index, each dimension is considered to have equal importance, and is thus given a $1/3$ weight in the overall framework. In turn, each indicator is given equal importance within dimensions. Therefore, by assigning weight w_j to the j-th indicator, we calculate a household-varying deprivation score, c_i, as following

$$c_i = \sum_{j=1}^{d} w_j g_i^0 \quad (1)$$

It is possible to build from the c-scores distribution to separate the population in two groups: the ones that face a critical number of weighted deprivations – these are the multidimensionally poor people – and the rest of the population. Identifying these two population subgroups is particularly relevant for our study as it allows us to determine the extent to which school closures reconfigure the proportion of the population living with a critical number of weighted deprivations, or the *size* of the poverty set (see Alkire and Foster, 2011). Following the UNDP-OPHI framework, the threshold to identify those people living in multidimensional poverty is $1/3$, i.e. the equivalent of one dimension. Thus, people having a c-score of $1/3$ or more are considered to be part of the poverty set.

Following this method, we construct three aggregate measures to have a sense of the magnitude and situation of people living with a critical number of weighted deprivations pre-school closures. Then, we will assess how these measures may be affected by school closures. The first aggregate measure is the *incidence* of multidimensional poverty, or the size of the poverty set. Following Alkire and Foster (2011), this incidence measure is denoted as H and is computed as the number of people in the poverty set, q, over the total population size, n: $H = q/n$. The second aggregate measure lays focus on the living conditions of the people belonging to the poverty set. This measure is termed poverty *intensity*, it is denoted as A, and is computed as the average c-score among the multidimensionally poor people: $A = E[c_i|c_i \geq 1/3]$. Finally, the third measure is the combination of the two previous ones; it is a measure of poverty incidence adjusted for intensity. It is denoted as M, and is computed as $M = H \times A = \left(\frac{q}{n}\right) E[c_i|c_i \geq 1/3]$. In our case, this measure has the advantage of capturing how school closure can shift the magnitude of people living with a critical number of deprivations induced by school closures, as well as the average deprivation suffered by those who were already poor.

Table 2 shows the current structure of the global MPI, which is computed following the AF methodology as we have just described it (see Alkire et al., 2022; Alkire & Santos, 2014). This index is formed of ten indicators regrouped in three dimensions, namely health, education and living standards. Each dimension is assigned an equal weight (1/3), and indicators are assigned equal weights within dimensions. A person is then identified as being multidimensionally poor if they simultaneously suffer 1/3 of the weighted deprivations or more.

Table 2: The global MPI structure

Dimension	Indicator	Deprived if…	Weight
Health (total weight: $\frac{1}{3}$)	Nutrition	Any adults have low BMI or persons aged 5 to 19 have low BMI-for-age or any child under 5 is underweight or stunted. *	$\frac{1}{6}$
	Child Mortality	Any child under 18 has died in the household in the five-year period preceding the survey. **	$\frac{1}{6}$
Education (total weight: $\frac{1}{3}$)	Years of schooling	No household member aged 12 years or older has completed six years of schooling.	$\frac{1}{6}$
	School attendance	Any school-aged child is not attending school up to the age at which he/she would complete class 8.	$\frac{1}{6}$
Living Standards (total weight: $\frac{1}{3}$)	Cooking fuel	The household cooks with dung, wood or charcoal.	$\frac{1}{18}$
	Sanitation	The household's sanitation facility is not improved, or it is improved but shared with other households.	$\frac{1}{18}$
	Drinking water	The household does not have access to improved drinking water or safe drinking water is at least a 30-minute walk from home, roundtrip.	$\frac{1}{18}$
	Electricity	The household has no electricity.	$\frac{1}{18}$

Housing	The household has inadequate housing: the floor is of natural materials or the roof or walls are of rudimentary materials.	$\frac{1}{18}$
Assets	The household does not own more than one of these assets: radio, TV, telephone, computer, animal cart, bicycle, motorbike, or refrigerator, and does not own a car or truck.	$\frac{1}{18}$

Source. Alkire, S., Kanagaratnam, U., Nogales, R., & Suppa, N. (2022).

*Regularly, the gMPI estimation includes nutrition as an indicator, however, the data used in this study does not contain information on nutrition measures. Thus, following OPHI's 51st Methodological Note, we do not consider that indicator, and we re-adjust the other health indicator weights (Alkire et al., 2021b).

**As the databases used in this study do not contain information on the age at which the child died or the date of his death, this indicator is defined as the number of dead children reported by each eligible woman in the household.

As we have stated earlier, the reason for using this particular structure of a multidimensional poverty index is that it includes school attendance as one of the two education indicators (with 1/6 weight in the overall structure). Using 2019 microdata, in Table 3 we show the proportion of people that live in households that experience school attendance deprivation as defined in the global MPI structure, as well as the interconnection of this deprivation with the other indicators that are considered in the index.

Table 3. Incidence of school attendance deprivation and its interconnection with other deprivations (2019)

Number of additional deprivations	**Mean**	**95% CI**
	%	
0	17.37	(8.86, 25.87)
1	15.51	(7.51, 23.51)
2	7.67	(2.95, 12.39)
3	11.74	(0.82, 22.66)
4	21.55	(9.66, 33.44)
5	14.10	(6.40, 21.81)
6	7.35	(3.11, 11.60)
7	3.67	(0.68, 6.66)
8	1.04	(-0.49, 2.57)

Note. The percentages sum up to 100% of the deprived population in school attendance. For this estimation, we use the sample size available in the 2019 survey for the deprived population in school attendance, i.e. 672 observations.

Clearly, if we were to choose one random observation from the 2019 dataset, that observation is more likely to be deprived in at least four additional indicators. Therefore, from the information presented in Table 3, we can conclude that facing school attendance problems is not expected to be the only livelihood struggle that many households have to endure on a daily

basis. Subsequently, Table 4 presents the estimated gMPI and its related indices from 2013 to 2019 at the national level.

Table 4: Estimation results for the gMPI, Bolivia 2013-2019

Year	MPI	95% CI	H	95% CI	A	95% CI	Uncensored headcount ratio	95% CI	Censored headcount ratio	95% CI
	Value		%		%		%		%	
2013	0.123	(0.112, 0.135)	25.3	(23.3, 27.2)	48.8	(47.4, 50.2)	43.4	(41.2, 45.7)	14.7	(12.8, 16.5)
2014	0.091	(0.083, 0.100)	20.0	(18.4, 21.6)	45.7	(44.7, 46.8)	4.1	(3.2, 5.0)	2.8	(2.0, 3.6)
2015	0.073	(0.066, 0.080)	16.9	(15.4, 18.3)	43.4	(42.6, 44.2)	2.9	(2.3, 3.6)	1.3	(0.9, 1.7)
2016	0.076	(0.068, 0.083)	16.7	(15.3, 18.1)	45.4	(44.1, 46.7)	2.9	(2.1, 3.7)	2.1	(1.4, 2.9)
2017	0.060	(0.054, 0.066)	13.8	(12.6, 15.1)	43.1	(42.2, 44.0)	0.7	(0.5, 1.0)	0.3	(0.1, 0.5)
2018	0.060	(0.054, 0.065)	13.8	(12.6, 14.9)	43.6	(43.6, 43.6)	1.5	(1.1, 1.9)	1.0	(0.6, 1.3)
2019	0.048	(0.043, 0.054)	11.5	(10.2, 12.7)	42.3	(41.5, 43.1)	2.1	(1.3, 3.0)	1.3	(0.5, 2.1)

Note. Table A2. in the appendix presents a full description of the estimated gMPIs and related indices from 2013 to 2019 at the national level and disaggregated at the subnational level.

3.2.2. **Projection methods**

In order to investigate the effects of school closures in the future, we need to identify the time-paths followed by i) school dropout rates and ii) measures of additional overlapping deprivations. With estimates of school attendance deprivation and multidimensional poverty measures available at just six points in time, it is not possible to apply traditional statistics-based approaches to identify the corresponding trajectories, or to forecast their levels by 2030 (such as ARIMA-class models). We overcome this data limitation by calibrating parametric trajectory models for these variables building on related work by Alkire et al. (2020). They posit the logistic model as the preferred specification for the dynamic behavior of multidimensional poverty aggregate measures based on the fact that they respect the variables' theoretical bounds (they are defined in the 0-1 range) and that there is strong cross-country evidence supporting it. Our strategy consists of taking the estimated values of the variables of interest denoted generically as $y(t)$, at the extreme points available in the time series (these are, $t = 2013, 2019$) to calibrate the parameters of the following logistic form:

$$\tilde{y}(t) = ln\,\frac{y(t)}{1-y(t)} = \alpha^{ln} - \beta^{ln} t \qquad (2)$$

where $y(t)$ represents any of the outcomes of interest. From the calibrated parameters, we then identify the trajectories following the logistic function of $y(t)$:

$$y(t) = \frac{1}{1+e^{-\alpha^{ln}+\beta^{ln} t}} \qquad (3)$$

The parameters obtained from the calibration process are presented in Appendix 4. We calibrate parameters at the national level and for each of the nine political subnational units (Chuquisaca, Cochabamba, Tarija, La Paz, Santa Cruz, Potosí, Oruro, Pando and Beni). Importantly, as a robustness analysis, we have also carried out statistical tests to corroborate the adequacy of our calibrated parameters – they can be found in Appendix 5). These tests consist in establishing whether the intermediate values observed for $y(t)$ from the available data (i.e. 2014-2018 data points) are indeed on the calibrated trajectory. This methodology allows us to make a prospective analysis of what could happen with i) school attendance rates and ii) overall poverty as measured by the experience of overlapping deprivations that make children become unavailable to return to school.

3.2.3. **Estimation of household-specific risk measures and school attendance shock simulations**

In order to assess the effect of school closures on their own, and to understand how they reconfigure the multiply deprived population sets in Bolivia, we conduct micro-simulations based on household-specific expected probabilities of becoming deprived in school attendance, as defined in the global MPI framework.

We assume that some households are more likely to become deprived in school attendance and/or are at greater risk of school absenteeism during the pandemic primarily due to lack of remote learning equipment. Thus, we first build a household-varying vulnerability score signaling effective access (or lack thereof) to virtual learning formats (see Brown et al., 2020). The scores build upon the latest available microdata (2019) on five key indicators of *lack of access* to technology such as no ownership of i) computer, ii) mobile phone iii) radio iv) television, and iv) lack of internet access by cable at home. Thus, the scores range from 0 to 5, where 0 represents the case in which the household presents no risk of school absenteeism (or total access to virtual learning formats), and 5 represents the case where a household has the highest risk of school absenteeism (or no access to virtual learning).

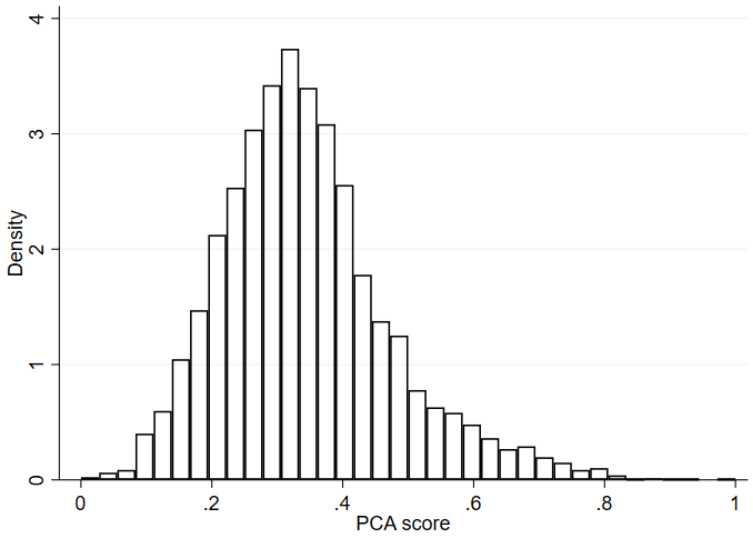
We argue that this vulnerability score is particularly useful to predict lack of access to virtual learning, but one must also consider other socioeconomic characteristics that are not directly related to technological access. Based on data availability, our additional predictors of lack of access to virtual learning include i) per capita income, ii) the c_i deprivation score deprivations defined earlier – as we expect risks of school unattendance to be correlated with deprivations in health, (parental) years of schooling, and household infrastructure (see Alkire et al., 2022) –, and iii) household composition, i.e., the number of children in the household.

The information contained in the vulnerability score and the three additional living condition indicators is summarized in a single continuous variable by applying a Principal Component Analysis (PCA). In this setting, the first principal component scores represent can be interpreted as a continuous representation of household-specific school unattendance risk during the pandemic. Re-scaling these PCA scores into the 0-1 range allows us to establish a household

partial ordering where 0 represents households having the least risk of school unattendance, and 1 represents those facing the highest risk.

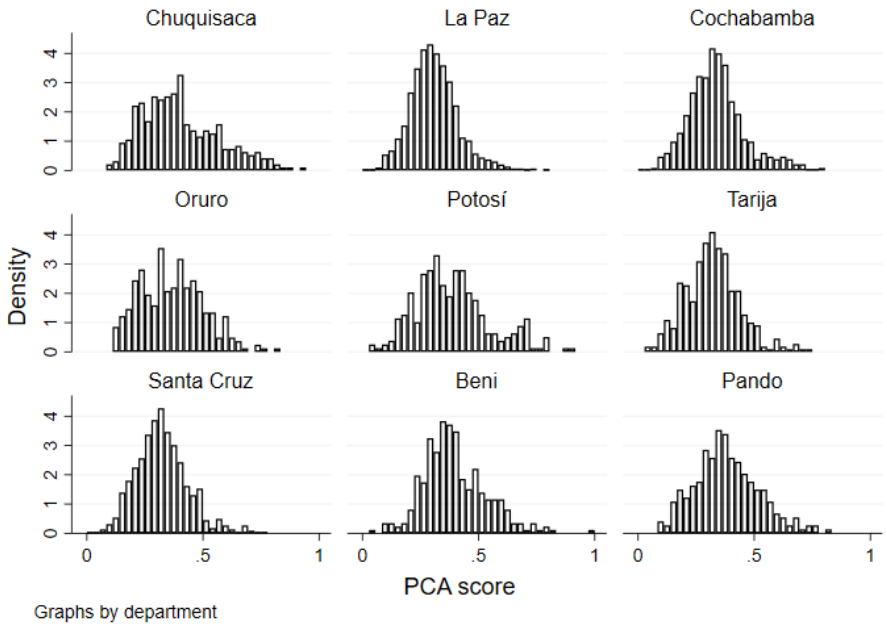
In Figure 1 we plot the distribution of the PCA scores, which is clearly right-skewed. The median score is 0.330 – this means that 50% of eligible households have a probability of becoming deprived in school attendance that is over one-third. Households sitting on the fourth quartile of this distribution – the most likely to become deprived in school attendance – have a deprivation probability of 41% or more.

Figure 1: Principal components index distribution



In Figure 2, we can see that the positive skewness of the national PCA score distribution is also found subnationally.

Figure 2: Principal components index distribution
(at the subnational level)



Note. This graph represents the distribution of the risk to be deprived in school attendance that Bolivian households present by department (at the subnational level). The index is obtained only for eligible households.

These PCA scores feed a Monte Carlo framework in order to simulate additional school unattendance deprivation among the vulnerable population. Our approach consists of three steps:

- First, we identify all children in the 2019 Household Survey that were attending school pre-pandemic. They are the population susceptible of acquiring the deprivation of interest in this study; they are the ones at risk of school absenteeism during the pandemic.
- Second, for every child in an at-risk household, we generate a thousand random draws from a Binomial distribution where the probability of becoming deprived is given by the household specific PCA score.
- Finally, the average across the results of the random draws for each household can be interpreted as the estimated probability of a non-deprived household to become deprived in school attendance.

The simulation procedure is carried out solely for eligible households, that is those households that are not deprived in school attendance as defined in the global MPI context and have at least one child regularly attending school in 2019. The likelihood of eligible households to become deprived in school attendance is defined as a direct function of the following predictors: i) the vulnerability score. ii) per capita household income, iii) the weighted deprivation count, c_i, and,

iv) the number of children in the household. We then simulated school attendance shocks under five scenarios:

- Scenario 1: the 10% most likely households effectively become deprived in school attendance
- Scenario 2: the 20% most likely households effectively become deprived in school attendance
- Scenario 3: the 30% most likely households effectively become deprived in school attendance
- Scenario 4: the 40% most likely households effectively become deprived in school attendance
- Scenario 5: the 50% most likely households effectively become deprived in school attendance

A schematic representation of this simulation procedure can be found in Appendix 1.

4. Results

Let us begin the discussion of our results by focusing on the potential effect of school closures on deprivations in school attendance as defined in the global MPI context. We analyze the possible consequences of having some households that prove unable to keep their children in school during and after school closures due to the pandemic. We first evaluate the potential increase in school attendance deprivations alone, and then we show the extent to which these simulated additional deprivations reshape the joint distribution of the ten livelihood indicators included in the global MPI.

Table 5 shows the baseline (observed) proportion of people living in households where there is at least one school-aged child not attending school in 2019. In this baseline scenario, it is clear that the COVID-19 pandemic took hold in Bolivia when school unattendance rates were quite low, in general. Around 2% of the total population lived in a household where there were school attendance problems, and this rate was under 4% in eight of the nine subnational regions (Chuquisaca is the exception); it was even under 1% in La Paz, Cochabamba, and Oruro.

Table 5. Actual and simulated incidence of school attendance deprivations (%)

Scenario	National	Chuquisaca	La Paz	Cochabamba	Oruro	Potosi	Tarija	Santa Cruz	Beni	Pando
Baseline	2.1	4.2	0.8	0.8	0.6	2.4	2.3	3.8	2.4	2.3
1	8.5	18.2	5.6	8.6	8.5	13.4	5.6	7.3	13	14.2
2	14.8	23.8	10.7	14	15.9	23.1	12.1	13.5	23.4	24.4
3	20.9	29.9	16.6	20.3	21.4	29.6	19.2	19	32.8	32.5
4	25.9	33.8	21	26.1	25.8	33.9	26.1	24	37.8	40.3
5	31.8	39.6	27	33	29.5	37.3	31.8	30.1	45.4	45.2

We find that school closures may have easily increased the observed incidence rates of deprivations school attendance by four times. In scenario 1, where the 10% of households that are most likely to become deprived in school attendance become effectively deprived, the proportion of people living in households where there is at least one school-aged child out of school would have gone up to 8.5%. It could have even gone up to 13.4% in Potosí, 14.2% in Pando, and 18.2% in Chuquisaca. This goes on to show how ill-prepared were the most vulnerable to adapt to virtual learning formats. In our worst-case scenario (5), where the 50% most vulnerable households become deprived in school attendance, the proportion of people living in households with at least one school-aged child out of school would have gone up to nearly 32% nationally, and it would have been over 45% in Beni and Pando.

We find that becoming deprived in school attendance reconfigures the households' entire deprivation profile. The simulated additional deprivations in school attendance combine themselves with pre-existing livelihood deprivations in other indicators included in the global MPI in such a way that they can push a considerable number of people into multidimensional poverty. As shown in Table 6, we find that the proportion of people in multidimensional poverty pre-pandemic was 11.5% nationally, and it went up to nearly 24% in Chuquisaca and 27% in Potosí. Under scenario 1 (10% of the most likely households become deprived in school attendance), the proportion of people in multidimensional poverty would have gone up to more than 14% nationally, and over 30% in Potosí. In our worst-case scenario (5, where 50% of the most likely population become deprived in school attendance), the incidence of multidimensional poverty would have gone up to 19% nationally, and up to 32% in Chuquisaca, and 38% in Potosí.

Table 6. Actual and simulated multidimensional poverty measures

	Scenario	National	Chuquisaca	La Paz	Cochabamba	Oruro	Potosi	Tarija	Santa Cruz	Beni	Pando
H *(%)*	**Baseline**	11.5	23.8	8.6	10.8	13.7	26.6	9.4	8.2	9.3	10.9
	1	14.1	28	10.7	13.8	16.6	30.4	9.7	10.3	16.1	15.8
	2	16.3	30.5	12.7	16.2	19.9	33.4	12.2	11.6	19.9	20.3
	3	18	32	14.1	18.4	21.2	36.1	14.9	12.6	22.9	22.7
	4	18.4	32	14.2	19.2	21.8	36.6	15.1	13.2	23.1	23.7
	5	19	32.2	14.7	20.4	22.3	37.7	15.3	13.3	24.2	23.9
M *(Value)*	**Baseline**	0.048	0.107	0.035	0.047	0.057	0.122	0.039	0.032	0.04	0.045
	1	0.064	0.139	0.046	0.066	0.076	0.148	0.044	0.042	0.07	0.073
	2	0.074	0.151	0.054	0.078	0.089	0.164	0.056	0.049	0.09	0.092
	3	0.081	0.157	0.06	0.086	0.095	0.174	0.066	0.054	0.1	0.102
	4	0.083	0.157	0.061	0.089	0.098	0.176	0.066	0.056	0.1	0.106
	5	0.085	0.158	0.063	0.094	0.099	0.18	0.067	0.057	0.1	0.107

Moreover, becoming deprived in school attendance not only implies falling into poverty for some people, but it can also imply exacerbating the experience of poverty among those who were already poor. In part, this can be seen in the increase of the adjusted incidence of multidimensional poverty (or the MPI value), which is also presented in Table 6. In our worst-case scenario (5), the value of the index would have almost doubled nationally, going from 0.048 to 0.085. In fact, if would have more than doubled in Beni (from 0.040 to 0.100) and Pando (from 0.045 to 0.107).

Our results clearly show that if the most vulnerable households were, indeed, unable to keep their children in school during or after school closures, then the newly acquired school attendance deprivation would become a long-lasting additional life struggle on top of pre-existing deprivations experienced by these households. The association between school unattendance and the experience of multidimensional poverty may be considerably strengthened if school closures due to the pandemic translate into school absenteeism. The extent of this problem is presented in Table 7, which shows the proportion of people that are multidimensionally poor and deprived in school attendance. In the baseline scenario, this part of the population is 1.3% nationally, and it is systematically equal or under 4% in each subnational region. However, in scenario 1 (10% most vulnerable become deprived in school attendance) the part of the population living in this particular situation would rise up to 7% nationally, and over 5% in almost every single region – Tarija being the exception with 3.5%. In our worst-case scenario (5), this proportion would be around 14% nationally, and systematically over 10% subnationally. In this scenario, the subnational regions with the largest part of their population living in multidimensional poverty and being deprived in school attendance would be Chuquisaca (~23%), Potosí (~23%), Beni (~22%), and Pando (~21%). Thus, these would be the most vulnerable subnational regions to the effects of the pandemic on school attendance.

Table 7: Proportion of people that are multidimensionally poor and deprived in school attendance

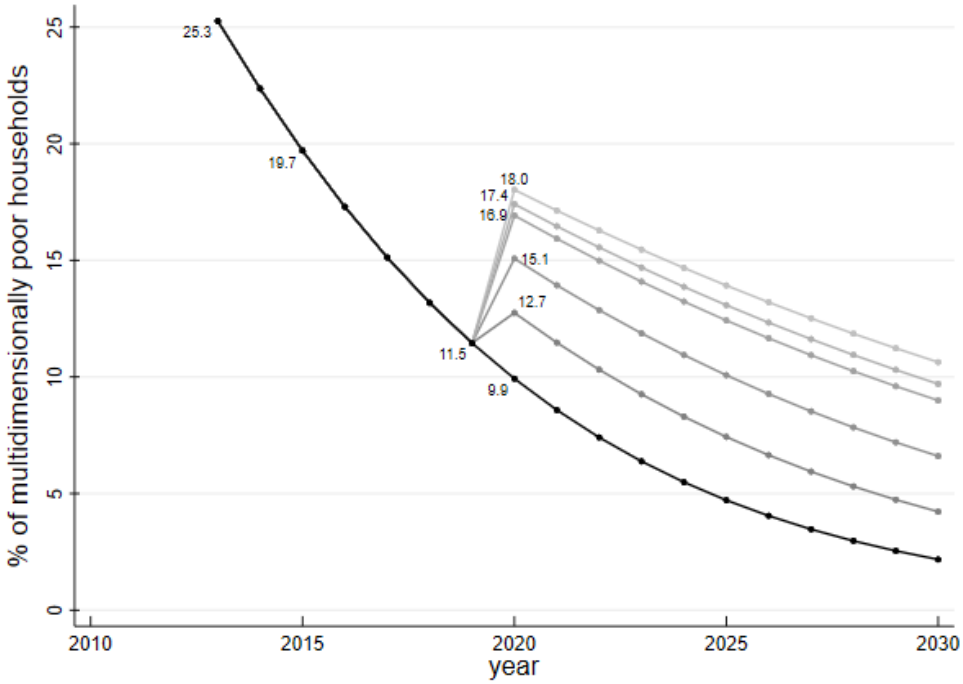
Scenario	National	Chuquisaca	La Paz	Cochabamba	Oruro	Potosi	Tarija	Santa Cruz	Beni	Pando
Baseline	1.3	4	0.5	0.3	0.3	2.1	1.2	2	1.9	0.7
1	7	16.8	4.5	7.5	7.4	12.7	3.5	5.1	11.4	10.8
2	10.5	20.4	7.4	11	11.9	18.9	7.4	8	16.8	16.7
3	12.5	22.4	8.9	13.6	13.8	21.7	10.4	9.5	19.9	19.8
4	13.2	22.6	9.4	14.6	14.7	22.6	10.6	10.2	20.2	21.2
5	13.9	23	10.1	15.9	15.2	23.6	11	10.6	21.5	21.4

Importantly, we find that if school closures due to the pandemic create persistent inabilities to keep children in school in the most vulnerable households, then hard-earned progress in terms of multidimensional poverty reductions (i.e. the experience of simultaneous overlapping deprivations) is at high risk of being erased. This situation can negatively affect processes of sustainable development in the country, increasing obstacles to meet development targets that would have been otherwise feasible if expected trends continues

To see this, let us now focus on the possible consequences of school closure-induced shifts in the trajectories of multidimensional poverty measures over time. Figure 3 depicts the expected business-as-usual incidence of multidimensional poverty until 2030 at the national level. We decided to project within this timeframe so that our results are informative of possible consequences on the 2030 Agenda for Sustainable Development. Our business-as-usual projections are a close proxy of Bolivia's pre-pandemic progress towards meeting target 1.2.2 established by the 2030 Agenda, which expresses the commitment of the international community to "reduce at least by half the proportion of men, women and children of all ages living in poverty in all its dimensions according to national definitions". Due to the lack of official measurement of multidimensional poverty in Bolivia, we make these assessments using the global MPI, following the procedure used by Alkire et al. (2020) described earlier.

The measured level of multidimensional poverty incidence was 19.7% for 2015, which means that the goal of halving the proportion of people living in multidimensional poverty required arriving at a 9.9% incidence in 2030. According to the expected business-as-usual trend in Figure 3 (black line), this target value would have, indeed, been achieved between 2019 and 2020. Thus, prior to the pandemic in 2020, Bolivia was on track to meet the goal of halving multidimensional poverty.

Figure 3: Expected national trends for the incidence of poverty



In addition to the business-as-usual trend of multidimensional poverty, in Figure 3 we also plot the trajectories followed by this rate in each of the simulated scenarios. Our results show that due to the shifts caused by the school attendance shock, poverty incidence could have risen from 9.9% (baseline) to 18% (scenario 5) in 2020, i.e. a rise of 8.1 percentage points. The detailed levels and shifts of multidimensional poverty incidence for 2020 at the national and subnational levels are presented in Table 8. Our results also show that the rise of poverty incidence from the baseline scenario to scenario 5 due to the shifts caused by the school attendance shock is over 9% in most subnational regions – with the exception of Santa Cruz (5.8 p.p.), La Paz (6.3 p.p.) and Tarija (6.5 p.p.). Also, the subnational regions with the largest increase of poverty incidence from the baseline to scenario 5 would be Beni (~16 p.p.), Pando (~14 p.p.) and Potosi (~12 p.p.). Of these regions, Potosi is the one that presents the highest level of poverty incidence in each scenario, which would go up to 36% in scenario 5. Clearly, the school closure-induced trajectory shifts can imply a considerable setback to progress made in terms of reducing multidimensional poverty across the entire country.

Additionally, in Table 8 we also present the response of the annual adjusted relative change of poverty incidence (or speed of transition), denoted as β. These rates of change decrease due to the school closure-induced shocks to the trajectories. Thus, not only do school closures imply a severe rise of multidimensional poverty in 2020, but they also imply lower rates of progress towards achieving SDG 1.2.2. At the national level this rate goes from its original value of 0.16 down to 0.06 in the worst-case scenario (scenario 5) at the national level. Decreases of these rates are also notable at the subnational level. These reductions are largest in the departments of Beni and Pando, which are some of the poorest regions subnationally by the global MPI. Thus,

we can conclude that the deprivation in school attendance caused by the pandemic is combined with another set of persistent deprivations that, even if held constant, can cause a considerable setback in the process of multidimensional poverty reduction due to the observed shock in school attendance. Solving school attendance problems, therefore, also involves solving other types of deprivations that also affect the most vulnerable households at the same time.

Table 8: Shock induced trajectory shifts for the incidence of poverty in 2020

Scenario	**pre-pandemic**		**1**			**2**			**3**			**4**			**5**		
	$\boldsymbol{H}$	$\boldsymbol{\beta}$	$\boldsymbol{H}$	$\boldsymbol{\beta}$	$\boldsymbol{\Delta^{H}_{1-pre}}$	$\boldsymbol{H}$	$\boldsymbol{\beta}$	$\boldsymbol{\Delta^{H}_{2-pre}}$	$\boldsymbol{H}$	$\boldsymbol{\beta}$	$\boldsymbol{\Delta^{H}_{3-pre}}$	$\boldsymbol{H}$	$\boldsymbol{\beta}$	$\boldsymbol{\Delta^{H}_{4-pre}}$	$\boldsymbol{H}$	$\boldsymbol{\beta}$	$\boldsymbol{\Delta^{H}_{5-pre}}$
National	9.9	0.16	12.7	0.12	2.8	15.1	0.09	5.1	16.9	0.07	7.0	17.4	0.07	7.5	18.0	0.06	8.1
Chuquisaca	20.9	0.17	25.4	0.13	4.5	28.1	0.11	7.3	29.8	0.10	9.0	29.9	0.10	9.1	30.1	0.10	9.3
La Paz	7.2	0.20	9.3	0.16	2.1	11.4	0.13	4.2	12.8	0.11	5.6	12.9	0.11	5.8	13.5	0.10	6.3
Cochabamba	9.0	0.20	12.1	0.15	3.0	14.6	0.12	5.6	17.0	0.10	8.0	17.9	0.09	8.9	19.2	0.07	10.2
Oruro	12.1	0.13	15.4	0.10	3.2	18.9	0.06	6.8	20.4	0.05	8.3	21.2	0.04	9.0	21.7	0.04	9.6
Potosi	23.6	0.16	27.8	0.13	4.1	31.1	0.10	7.5	34.2	0.08	10.6	34.8	0.08	11.2	36.0	0.07	12.4
Tarija	8.3	0.14	8.6	0.13	0.4	11.3	0.09	3.0	14.3	0.05	6.1	14.5	0.05	6.3	14.7	0.04	6.5
Santa Cruz	7.6	0.09	10.0	0.04	2.3	11.3	0.02	3.7	12.6	0.01	4.9	13.2	0.00	5.6	13.4	0.00	5.8
Beni	7.5	0.24	14.4	0.13	6.9	18.6	0.09	11.1	21.9	0.06	14.4	22.2	0.06	14.7	23.4	0.05	15.9
Pando	9.1	0.20	14.1	0.13	5.0	19.1	0.08	10.0	21.8	0.05	12.7	22.9	0.04	13.8	23.1	0.04	14.0

5. Concluding remarks

The COVID-19 pandemic has created unprecedented challenges for people around the world and for their governments. While closing schools may have indeed limited the spread of the virus and thus helped save lives, it is a measure that has come with costs. Precisely identifying these costs is not yet possible due to current lack of data for proper impact evaluations, a situation that is likely to remain unchanged in the short term. Meanwhile, by early 2022, school closures remained only partially open in some of the poorest countries, including Bolivia, with virtual learning modalities having been fostered for almost two years. The issue is that there is compelling evidence showing that children that are out of school for long periods of time, are likely to drop out subsequently (Azevedo et al., 2021). Clearly, even with partial information, policy action needs to take place, which is why the main focus of this paper is to generate objective evidence about possible consequences of school closures on school unattendance, and how this situation may interact with pre-existing livelihood struggles including poor health, low parental education, and housing deprivations.

Our analysis emphasizes that many Bolivian learners lacked internet connectivity and technological devices, meaning that they were effectively out of school. These technological deprivations are endured simultaneously with other deprivations -in the domains of health, education, and living standards. We show that in these circumstances, if 10% of children of these vulnerable households become effectively deprived in school attendance, the proportion of people living in households where there is at least one school-aged child out of school would have gone up from 2.1% to 8.5%. In fact, this is a lower bound.

While the potential increase of school unattendance in Bolivia already calls for urgent policy action on its own, we wish to stress that consequences are even more far-reaching. For the most vulnerable households, school attendance deprivation implies enduring one additional life struggle that may push them into multidimensional poverty. If 10% of children in the most vulnerable households become deprived in school attendance, the proportion of people in multidimensional poverty would have gone up from 11.5% pre-pandemic to more than 14% nationally. The proportion of people that are multidimensionally poor and deprived in school attendance would then go up from 1.3% to 7%. Again, these figures depict lower bounds.

Before the pandemic, Bolivia was well on track to achieve universal school attendance, and to reduce by half the proportion of men, women, and children living in multidimensional poverty, as stated in the 2030 Agenda for Sustainable Development. In fact, this ambitious poverty target was projected to be met by 2020. School closures may have postponed achievement of this goal by two to three years. This goes on to show the extent to which processes of sustainable development have been affected by school closures in the country. We reiterate the usefulness of this drastic measure in some regards, chiefly related to containing the spread of the virus at a crucial time. However, it is now high time that governments started designing strategies for

recovery post-pandemic. The information presented in our study may thus be a useful input for this urgent conversation in the policy sphere. Naturally, in academic spheres, our study may also be used as input for discussions methods to overcome evident data limitations and deficiencies in order to objectively inform urgent policymaking to prevent development progress from being erased. Otherwise, the most vulnerable will be the first to pay the consequences of either policy inaction or inadequate policymaking decisions. This would clearly go against the pledge of leaving no one behind in the process toward sustainable development, which is so powerfully pitched in virtually all development agendas.

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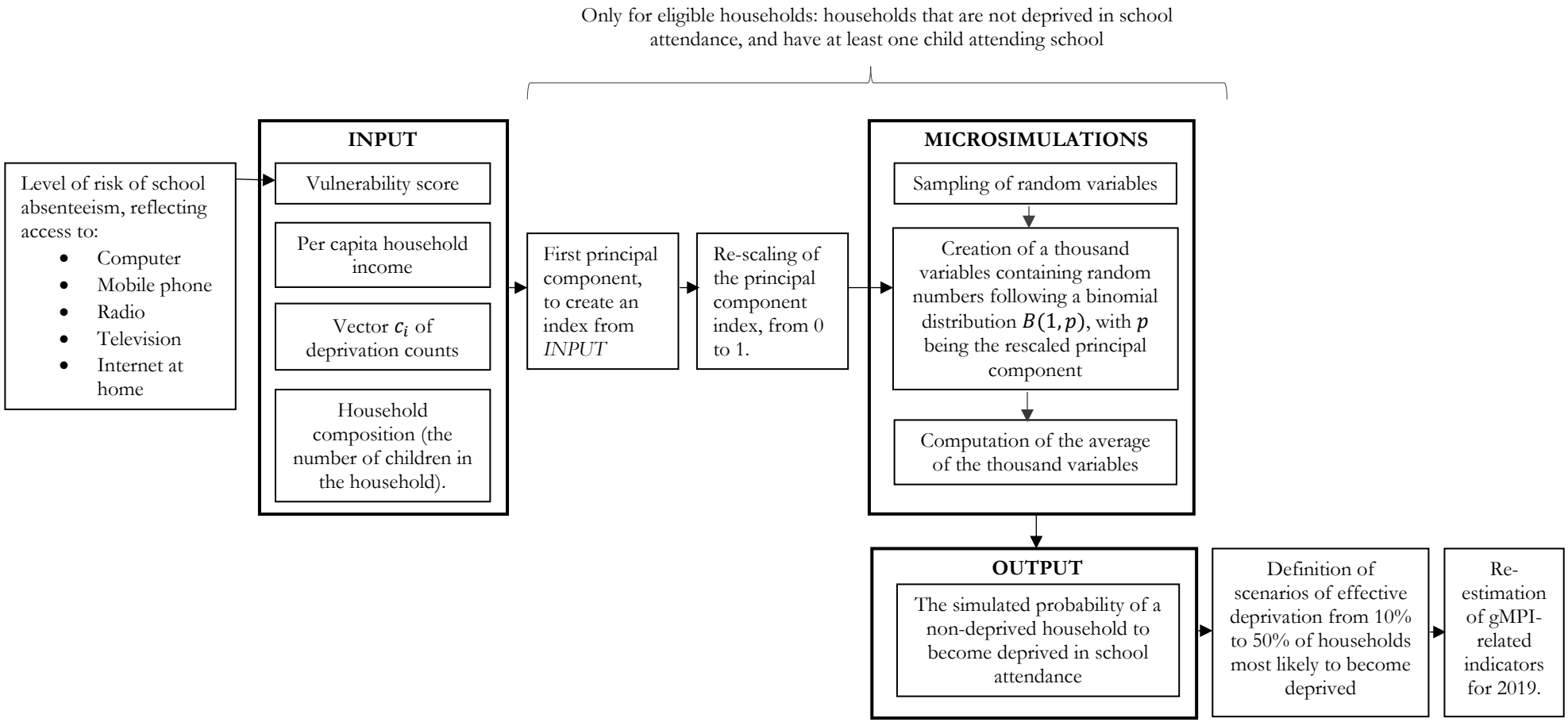
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7. Appendices

7.1. Appendix 1. Microsimulation's procedure diagram



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7.2. Appendix 2. Relevant variables

In Table A1 we show some information about relevant variables in these datasets – they correspond to the frequency and percentage of households reporting different deprivations in 2013 to 2019. Initially, we consider information on child mortality deprivations (first row). In 2013, 7.8% of households in Bolivia reported the death of at least one child, while in 2019, only 3.9% of households reported the same thing. Similarly, we include information on deprivations on schooling. In 2013, 16.3% of households in Bolivia reported having no members over 12 who completed six years of schooling, while in 2019, 14.9% of households reported the same thing. For both variables, there is a slight but remarkable decrease of households reporting deprivation within the same period. Then, we observe information on some living standards deprivations, including cooking fuel, sanitation, drinking water, electricity, housing, and assets. For cooking fuel, sanitation, electricity and housing deprivations there are important decreases between 2013 and 2019. The decrease observed for households reporting cooking with any solid fuel such as dung, wood or charcoal is about 7.9 percentage points. There is a decrease of 8.5 percentage points of households that reported having a sanitation facility not improved, or improved but shared with other households. The decrease for households reporting having no electricity is about 5.2 percentage points. The percentage of households that reported having inadequate housing presents the biggest decrease of about 15.5 percentage points. On the other hand, drinking water and assets deprivations presented small decreases or even increases between the same period. The decrease observed for households reporting owning no more than one of the selected assets including radio, TV, computer, bicycle, motorcycle or refrigerator), and not owning a car or truck is only about 0.8 percentage points. Lastly, the percentage of households that reported having no access to improved drinking water presents an increase of about 7.9 percentage points between 2013 and 2019.

The most important variable considered for this study can be found in the first row of Table A1, and it describes households that reported having at least one school-aged child not attending school (i.e. school attendance deprivations). As can be noticed, there is a marked decrease in the percentage of households reporting school attendance deprivations between 2013 and 2014, of about 29.3 percentage points. Then, it remains relatively stable until 2019, showing a decrease of only 1.3 percentage points between 2014 and 2019.

Table A1: Description of relevant variables (in thousands)

	2013		2014		2015		2016		2017		2018		2019	
Households reporting…	**N**	**%**	**N**	**%**	**N**	**%**	**N**	**%**	**N**	**%**	**N**	**%**	**N**	**%**
at least one school-aged child not attending school.*	894	31.9	75	2.6	57	1.9	55	1.7	15	0.5	33	1.0	43	1.3
the death of at least one child.	219	7.8	220	7.6	174	5.8	183	5.7	141	4.2	153	4.4	136	3.9
no members over 12 who have completed six years of schooling.	456	16.3	537	18.7	491	16.4	541	16.8	610	18.2	565	16.3	513	14.9
cooking with any solid fuel such as dung, wood or charcoal.	551	19.6	597	20.7	526	17.5	523	16.3	520	15.5	524	15.1	407	11.8
a sanitation facility not improved, or improved but shared with other households.	1,286	45.8	1,465	50.9	1,318	43.8	1,424	44.4	1,413	42.2	1,431	41.3	1,288	37.3
no access to improved drinking water.	767	27.3	768	26.7	887	29.5	1,055	32.9	962	28.8	1,093	31.6	1,216	35.2
no electricity.	294	10.5	287	10.0	255	8.5	271	8.4	272	8.1	250	7.2	182	5.3
inadequate housing.**	1,272	45.3	1,267	44.0	1,208	40.1	1,202	37.5	1,260	37.7	1,250	36.1	1,032	29.9
owning no more than one of the selected assets***, and not owning a car or truck.	646	23.1	692	24.0	715	23.8	824	25.7	807	24.1	801	23.1	769	22.3

Notes. N is obtained by weighting the number of observations in the survey by its expansion factor. *Up to age at which he/she would complete class 8: 14 years. **The floor is of natural materials, or the roof or walls are of rudimentary materials. ***Radio, TV, computer, bicycle, motorcycle or refrigerator

7.3. Appendix 3. Estimated MPIs

Table A2. Estimated gMPIs and related indices (2013-2019)

Department	Year	gMPI	c.i.		*H*	c.i.		*A*	c.i.	
		Value			*%*			*%*		
National	**2013**	0.123	(0.112,	0.135)	25.3	(23.3,	27.2)	48.8	(47.4,	50.2)
	2014	0.091	(0.083,	0.100)	20.0	(18.4,	21.6)	45.7	(44.7,	46.8)
	2015	0.073	(0.066,	0.080)	16.9	(15.4,	18.3)	43.4	(42.6,	44.2)
	2016	0.076	(0.068,	0.083)	16.7	(15.3,	18.1)	45.4	(44.1,	46.7)
	2017	0.060	(0.054,	0.066)	13.8	(12.6,	15.1)	43.1	(42.2,	44.0)
	2018	0.060	(0.054,	0.065)	13.8	(12.6,	14.9)	43.6	(43.6,	43.6)
	2019	0.048	(0.043,	0.054)	11.5	(10.2,	12.7)	42.3	(41.5,	43.1)
Chuquisaca	**2013**	0.245	(0.245,	0.245)	46.2	(46.2,	46.2)	53.0	(53.0,	53.0)
	2014	0.137	(0.107,	0.167)	31.2	(25.2,	37.1)	43.9	(43.9,	43.9)
	2015	0.172	(0.132,	0.213)	38.2	(30.2,	46.3)	45.1	(45.1,	45.1)
	2016	0.138	(0.112,	0.164)	30.0	(24.9,	35.1)	46.1	(44.1,	48.1)
	2017	0.098	(0.070,	0.126)	23.1	(16.2,	29.9)	42.3	(42.3,	42.3)
	2018	0.125	(0.082,	0.168)	27.2	(18.4,	36.1)	45.9	(45.9,	45.9)
	2019	0.107	(0.077,	0.138)	23.8	(17.4,	30.1)	45.2	(43.7,	46.7)
La Paz	**2013**	0.114	(0.114,	0.114)	24.1	(24.1,	24.1)	47.2	(47.2,	47.2)
	2014	0.088	(0.074,	0.103)	19.9	(17.1,	22.7)	44.4	(42.7,	46.1)
	2015	0.066	(0.056,	0.076)	15.6	(13.5,	17.7)	42.2	(40.8,	43.6)
	2016	0.065	(0.052,	0.077)	14.7	(12.0,	17.4)	44.0	(42.2,	45.8)
	2017	0.055	(0.045,	0.066)	13.2	(10.9,	15.4)	42.0	(40.0,	43.9)
	2018	0.051	(0.043,	0.060)	12.4	(10.4,	14.4)	41.4	(39.5,	43.4)
	2019	0.035	(0.025,	0.044)	8.6	(6.4,	10.9)	40.3	(39.0,	41.6)
Cochabamba	**2013**	0.136	(0.136,	0.136)	28.5	(28.5,	28.5)	47.9	(47.9,	47.9)
	2014	0.079	(0.058,	0.099)	17.9	(13.8,	22.0)	44.1	(41.5,	46.7)
	2015	0.071	(0.058,	0.084)	16.5	(13.7,	19.4)	42.8	(41.2,	44.4)
	2016	0.087	(0.065,	0.108)	17.9	(14.8,	21.1)	48.5	(43.4,	53.5)
	2017	0.058	(0.047,	0.069)	13.0	(10.8,	15.2)	44.3	(42.4,	46.2)
	2018	0.066	(0.055,	0.077)	14.8	(12.5,	17.2)	44.4	(42.8,	46.1)
	2019	0.047	(0.037,	0.057)	10.8	(8.5,	13.0)	43.4	(41.8,	45.1)
Oruro	**2013**	0.136	(0.136,	0.136)	26.2	(26.2,	26.2)	51.7	(51.7,	51.7)
	2014	0.087	(0.062,	0.112)	19.8	(14.4,	25.1)	43.8	(43.8,	43.8)
	2015	0.086	(0.064,	0.108)	19.3	(14.7,	24.0)	44.5	(44.5,	44.5)
	2016	0.074	(0.060,	0.087)	17.8	(14.7,	20.9)	41.4	(40.0,	42.8)
	2017	0.059	(0.045,	0.074)	14.3	(10.6,	17.9)	41.7	(40.3,	43.2)
	2018	0.079	(0.053,	0.105)	18.5	(13.2,	23.9)	42.6	(40.4,	44.8)
	2019	0.057	(0.043,	0.070)	13.7	(10.6,	16.7)	41.5	(39.4,	43.6)
Potosi	**2013**	0.251	(0.251,	0.251)	48.3	(48.3,	48.3)	52.1	(52.1,	52.1)

	2014	0.221	(0.185,	0.256)	42.9	(36.3,	49.4)	51.5	(48.8,	54.2)
	2015	0.161	(0.131,	0.192)	35.7	(28.9,	42.5)	45.2	(43.7,	46.7)
	2016	0.181	(0.146,	0.216)	37.0	(30.9,	43.0)	49.0	(46.6,	51.4)
	2017	0.165	(0.139,	0.192)	37.1	(31.0,	43.1)	44.6	(43.4,	45.8)
	2018	0.160	(0.129,	0.191)	34.5	(28.6,	40.4)	46.5	(44.0,	48.9)
	2019	0.122	(0.093,	0.150)	26.6	(20.6,	32.5)	45.8	(43.9,	47.6)
Tarija	**2013**	0.088	(0.088,	0.088)	19.1	(19.1,	19.1)	46.3	(46.3,	46.3)
	2014	0.056	(0.041,	0.071)	14.0	(10.2,	17.7)	40.2	(40.2,	40.2)
	2015	0.049	(0.037,	0.061)	11.6	(8.9,	14.3)	42.0	(42.0,	42.0)
	2016	0.054	(0.041,	0.067)	12.4	(9.6,	15.2)	43.5	(40.3,	46.7)
	2017	0.047	(0.029,	0.065)	11.1	(7.8,	14.4)	42.1	(42.1,	42.1)
	2018	0.049	(0.032,	0.066)	11.1	(7.8,	14.5)	43.8	(40.6,	47.0)
	2019	0.039	(0.013,	0.066)	9.4	(3.8,	14.9)	41.8	(41.8,	41.8)
Santa Cruz	**2013**	0.063	(0.045,	0.081)	13.0	(10.0,	16.0)	48.2	(48.2,	48.2)
	2014	0.063	(0.050,	0.077)	13.9	(11.3,	16.5)	45.3	(42.7,	47.8)
	2015	0.039	(0.031,	0.046)	9.3	(7.7,	11.0)	41.3	(39.3,	43.3)
	2016	0.040	(0.030	0.050)	9.6	(7.4,	11.8)	41.6	(39.7,	43.4)
	2017	0.032	(0.024,	0.040)	7.5	(5.7,	9.3)	42.9	(39.6,	46.1)
	2018	0.026	(0.019,	0.032)	6.4	(4.9,	7.9)	40.2	(37.9,	42.5)
	2019	0.032	(0.023,	0.041)	8.2	(5.8,	10.7)	38.9	(37.7,	40.0)
Beni	**2013**	0.124	(0.124,	0.124)	29.5	(29.5,	29.5)	41.8	(39.4,	44.2)
	2014	0.082	(0.046,	0.117)	17.0	(24.1,	9.9)	48.2	(48.2,	48.2)
	2015	0.068	(0.039,	0.097)	13.9	(18.8,	9.1)	48.8	(48.8,	48.8)
	2016	0.084	(0.050,	0.118)	19.2	(26.1,	12.2)	44.0	(40.1,	47.8)
	2017	0.047	(0.033,	0.061)	11.3	(14.5,	8.2)	41.9	(40.3,	43.5)
	2018	0.047	(0.029,	0.065)	10.8	(14.3,	7.2)	43.7	(40.5,	46.9)
	2019	0.039	(0.024,	0.054)	9.3	(12.3,	6.2)	42.2	(42.2,	42.2)
Pando	**2013**	0.134	(0.105,	0.163)	28.8	(22.7,	35.0)	46.6	(43.2,	50.1)
	2014	0.097	(0.072,	0.121)	20.2	(15.2,	25.2)	47.8	(45.2,	50.4)
	2015	0.085	(0.061,	0.109)	19.3	(13.8,	24.8)	44.1	(41.5,	46.6)
	2016	0.062	(0.042,	0.083)	14.3	(9.5,	19.1)	43.5	(43.5,	43.5)
	2017	0.056	(0.037,	0.076)	12.6	(8.0,	17.3)	44.6	(44.6,	44.6)
	2018	0.037	(0.022,	0.052)	8.5	(5.4,	11.6)	43.9	(43.9,	43.9)
	2019	0.045	(0.029,	0.062)	10.9	(6.9,	14.9)	41.5	(41.5,	41.5)

7.4. Appendix 4. Calibrated parameters

The parameters obtained from the calibration process are presented in Table A3. We calibrate parameters at the national level and for each of the nine political subnational units (Chuquisaca, Cochabamba, Tarija, La Paz, Santa Cruz, Potosí, Oruro, Pando and Beni).

Table A3: Calibrated parameters, Bolivia

		National	**Chuquisaca**	**La Paz**	**Cochabamba**
M	α	339.05	330.96	424.79	390.19
	β	0.17	0.16	0.21	0.19
H	α	321.23	339.09	406.39	398.99
	β	0.16	0.17	0.20	0.20

		Potosí	**Tarija**	**Santa Cruz**	**Beni**	**Pando**
M	α	295.91	287.90	234.05	414.48	396.58
	β	0.15	0.14	0.12	0.21	0.20
H	α	317.33	275.39	169.33	472.21	400.91
	β	0.16	0.14	0.09	0.24	0.20

Note. The parameters presented here are calibrated assuming a pre-pandemic scenario, and they do not consider any shock on school attendance.

7.5. Appendix 5. Statistical tests

These tests consist in establishing whether the intermediate values observed for $y(t)$ from the available data (i.e. 2014-2018 data points) are indeed on the calibrated trajectory.

Table A4 and Table A5 present statistical tests that are performed at the national and subnational levels to corroborate that the intermediate values observed for $H(t)$ and $M(t)$ (respectively) from the available data are on the calibrated trajectory obtained by the process described in section 3.2.2., in order to test the validation of the model and the adequacy of the identified trajectories. The null hypothesis evaluated states that the values obtained through the calibrated logistic model are statistically equal to the observed values at 1% significance. Most of the tests conclude not to reject the null hypothesis. For those cases which the adequacy is not met and the null hypothesis is rejected, the p-value is highlighted.

Table A4. Statistical testing of calibrated incidence H (2014-2018)

Department	**Year**	**Observed H**	**S.E.**	**Calibrated H**	**T-statistic**	**p-value**
National	2014	20.0	0.832	22.4	2.8	**0.0052**
	2015	16.9	0.734	19.7	3.9	**0.0020**
	2016	16.7	0.708	17.3	0.9	0.3682
	2017	13.8	0.657	15.1	1.9	0.0574

	2018	13.8	0.590	13.2	1.0	0.3174
Chuquisaca	2014	31.2	3.042	42.0	3.6	**0.0004**
	2015	38.2	4.127	38.0	0.1	0.9204
	2016	30.0	2.601	34.1	1.6	0.1320
	2017	23.1	3.475	30.4	2.1	0.0358
	2018	27.2	4.506	27.0	0.1	0.9204
La Paz	2014	19.9	1.434	20.6	0.5	0.6170
	2015	15.6	1.087	17.5	1.7	0.0892
	2016	14.7	1.367	14.8	0.0	1.0000
	2017	13.2	1.145	12.4	0.7	0.4840
	2018	12.4	1.009	10.4	2.0	0.0456
Cochabamba	2014	17.9	2.088	24.6	3.2	**0.0014**
	2015	16.5	1.463	21.1	3.1	**0.0020**
	2016	17.9	1.612	18.0	0.0	1.0000
	2017	13.0	1.112	15.2	2.0	0.0456
	2018	14.8	1.219	12.8	1.6	0.1096
Oruro	2014	19.8	2.715	23.7	1.4	0.1616
	2015	19.3	2.370	21.3	0.9	0.3682
	2016	17.8	1.585	19.2	0.9	0.3682
	2017	14.3	1.844	17.2	1.6	0.1096
	2018	18.5	2.745	15.3	1.2	0.2302
Potosí	2014	42.9	3.355	44.3	0.4	0.6892
	2015	35.7	3.476	40.5	1.4	0.1616
	2016	37.0	3.096	36.8	0.1	0.9204
	2017	37.1	3.093	33.2	1.3	0.1936
	2018	34.5	2.992	29.8	1.6	0.1096
Tarija	2014	14.0	1.926	17.0	1.6	0.1096
	2015	11.6	1.373	15.2	2.6	**0.0094**
	2016	12.4	1.416	13.5	0.8	0.4238
	2017	11.1	1.684	12.0	0.5	0.6170
	2018	11.1	1.707	10.6	0.3	0.7642
Santa Cruz	2014	13.9	1.322	12.1	1.4	0.1616
	2015	9.3	0.858	11.2	2.2	0.0278
	2016	9.6	1.118	10.4	0.7	0.4840
	2017	7.5	0.906	9.6	2.4	0.0164
	2018	6.4	0.777	8.9	3.2	**0.0014**
Beni	2014	17.0	3.628	24.9	2.2	0.0278
	2015	13.9	2.473	20.8	2.8	**0.0052**
	2016	19.2	3.531	17.2	0.6	0.5486
	2017	11.3	1.599	14.1	1.7	0.0892
	2018	10.8	1.815	11.5	0.4	0.6892

Pando	2014	20.2	2.556	24.9	1.8	0.0718
	2015	19.3	2.813	21.4	0.7	0.4840
	2016	14.3	2.447	18.2	1.6	0.1096
	2017	12.6	2.393	15.4	1.2	0.2302
	2018	8.5	1.576	13.0	2.9	**0.0038**

Table A5. Statistical testing of calibrated gMPI M (2014-2018)

Department	Year	Observed M	S.E.	Calibrated M	T-statistic	p-value
National	2014	0.091	0.004335	0.106	3.4	**0.0006**
	2015	0.073	0.003424	0.091	5.2	**0.0000**
	2016	0.076	0.003758	0.078	0.6	0.5486
	2017	0.060	0.003009	0.067	2.3	0.0214
	2018	0.060	0.002801	0.057	1.1	0.2714
Chuquisaca	2014	0.137	0.015160	0.215	5.2	**0.0000**
	2015	0.172	0.020813	0.189	0.8	0.4238
	2016	0.138	0.013205	0.165	2.0	0.0456
	2017	0.098	0.014284	0.143	3.2	**0.0014**
	2018	0.125	0.021993	0.124	0.0	1.0000
La Paz	2014	0.088	0.007239	0.094	0.8	0.4238
	2015	0.066	0.005035	0.078	2.3	0.0214
	2016	0.065	0.006306	0.064	0.1	0.9204
	2017	0.055	0.005325	0.052	0.6	0.5486
	2018	0.051	0.004444	0.043	2.0	0.0456
Cochabamba	2014	0.079	0.010524	0.115	3.4	**0.0006**
	2015	0.071	0.006511	0.097	4.0	**0.0000**
	2016	0.087	0.011027	0.081	0.5	0.6170
	2017	0.058	0.005630	0.068	1.7	0.0892
	2018	0.066	0.005403	0.056	1.8	0.0718
Oruro	2014	0.087	0.012706	0.118	2.5	0.0124
	2015	0.086	0.011259	0.102	1.4	0.1616
	2016	0.074	0.006773	0.088	2.2	0.0278
	2017	0.059	0.007525	0.076	2.2	0.0278
	2018	0.079	0.013171	0.066	1.0	0.3174
Potosí	2014	0.221	0.018215	0.225	0.2	0.8414
	2015	0.161	0.015468	0.200	2.5	0.0124
	2016	0.181	0.017966	0.177	0.2	0.8414
	2017	0.165	0.013539	0.157	0.6	0.5486
	2018	0.160	0.015782	0.138	1.4	0.1616
Tarija	2014	0.056	0.007737	0.077	2.7	**0.0070**

	2015	0.049	0.006094	0.068	3.1	**0.0020**
	2016	0.054	0.006678	0.059	0.8	0.4238
	2017	0.047	0.009008	0.052	0.5	0.6170
	2018	0.049	0.008742	0.045	0.4	0.6892
Santa Cruz	2014	0.063	0.006863	0.056	1.0	0.3174
	2015	0.039	0.003908	0.050	3.0	**0.0026**
	2016	0.040	0.004873	0.045	1.0	0.3174
	2017	0.032	0.004035	0.040	2.0	0.0456
	2018	0.026	0.003364	0.036	3.0	**0.0026**
Beni	2014	0.082	0.018095	0.103	1.2	0.2302
	2015	0.068	0.014984	0.085	1.2	0.2302
	2016	0.084	0.017310	0.070	0.8	0.4238
	2017	0.047	0.007164	0.058	1.5	0.1336
	2018	0.047	0.008973	0.048	0.1	0.9204
Pando	2014	0.097	0.012613	0.113	1.3	0.1936
	2015	0.085	0.012062	0.095	0.8	0.4238
	2016	0.062	0.010376	0.079	1.6	0.1096
	2017	0.056	0.009941	0.066	0.9	0.3682
	2018	0.037	0.007768	0.055	2.2	0.0278

Table A6 and Table A7 present statistical tests to corroborate that the calibrated parameters are statistically equal to the parameters obtained by a simple estimation. To obtain the data presented in the following tables, the following procedure was therefore performed:

- A simple regression was performed by department in the form $y(t) = \alpha + \beta t + e$, where $y(t)$ here is the logarithmic transformation of the variables of interest (the headcount ratio H, and the gMPI M).
- The estimators and standard errors of α and β are obtained at the national and the subnational levels.
- Statistical testing is performed to test null hypothesis that the estimated parameters are statistically equal to those that have been calibrated in section 2.2.2. The degrees of freedom in this case are the points in each regression minus two (i.e. seven minus two).

All tests concluded in not rejecting the null hypothesis.

Table A6. Statistical testing of calibrated parameters of H

Department	Parameter	Estimated	S.E.	Calibrated	T-statistic	p-value
National	β	0.14	0.016	0.16	1.0	0.3632
	α	287.20	32.912	321.23	1.0	0.3632
Chuquisaca	β	0.15	0.041	0.17	0.5	0.6382
	α	297.37	83.615	339.09	0.5	0.6382
La Paz	β	0.18	0.018	0.20	1.4	0.2204
	α	355.99	37.234	406.39	1.4	0.2204
Cochabamba	β	0.15	0.036	0.20	1.2	0.2838
	α	307.91	73.001	398.99	1.2	0.2838
Oruro	β	0.11	0.030	0.13	1.0	0.3632
	α	210.90	60.531	270.38	1.0	0.3632
Potosí	β	0.12	0.025	0.16	1.3	0.2504
	α	250.52	50.451	317.33	1.3	0.2504
Tarija	β	0.11	0.024	0.14	1.2	0.2838
	α	216.73	49.338	275.39	1.2	0.2838
Santa Cruz	β	0.12	0.033	0.09	1.2	0.2838
	α	249.47	66.490	169.33	1.2	0.2838
Beni	β	0.20	0.048	0.24	0.8	0.4600
	α	395.51	96.907	472.21	0.8	0.4600
Pando	β	0.22	0.034	0.20	0.5	0.6382
	α	438.09	69.464	400.91	0.5	0.6382

Table A7. Statistical testing of calibrated parameters of M

Department	Parameter	Estimated	S.E.	Calibrated	T-statistic	p-value
Nacional	β	0.14	0.018	0.17	1.8	0.1318
	α	270.26	35.972	339.05	1.9	0.1158
Chuquisaca	β	0.12	0.038	0.16	1.3	0.2504
	α	225.73	75.975	330.96	1.4	0.2204
La Paz	β	0.17	0.019	0.21	2.1	0.0898
	α	340.14	37.724	424.79	2.2	0.0790
Cochabamba	β	0.13	0.036	0.19	1.7	0.1498
	α	264.45	72.909	390.19	1.7	0.1498
Oruro	β	0.11	0.032	0.16	1.5	0.1940
	α	221.08	64.628	319.64	1.5	0.1940
Potosí	β	0.10	0.020	0.15	2.4	0.0616
	α	195.01	40.735	295.91	2.5	0.0544
Tarija	β	0.10	0.028	0.14	1.6	0.1704
	α	191.08	57.206	287.90	1.7	0.1498
Santa Cruz	β	0.14	0.031	0.12	0.8	0.4600
	α	280.21	62.340	234.05	0.7	0.5152
Beni	β	0.18	0.031	0.21	1.0	0.3632
	α	346.36	62.280	414.48	1.1	0.3214
Pando	β	0.20	0.026	0.20	0.1	0.9242
	α	395.27	52.831	396.58	0.0	1.0000